Understanding Drivers of National CO_2 Emissions

IE 7275: Data Mining in Engineering

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Problem Setting

Greenhouse gas emissions, specifically CO_2 generated through energy consumption, are the primary drivers of anthropogenic climate change. Differences in energy systems, economic development, and infrastructure, however, create significant disparities in national emissions profiles.

Addressing these disparities is at the heart of the United Nations Sustainable Development Goal 7 (SDG 7), which calls on all nations to "ensure affordable, reliable, sustainable, and modern energy for all" (United Nations, 2025) by 2030. The program is organized around three core targets: universal energy access (7.1), a substantial increase in the renewable share of the global energy mix (7.2), and doubling the rate of energy efficiency improvement (7.3). Despite measurable improvement from 2015 to 2023, progress has slowed. Approximately 666 million people still lack access to electricity, and 2.1 billion people still rely on polluting fuels (United Nations, 2025). Identifying the structural factors that differentiate successful transitions from stalled ones is therefore essential for designing effective climate policy.

Problem Definition

This project addresses two research questions:

1. What energy, economic, and infrastructure features most strongly predict a country's per-capita CO_2 emissions? How accurately can emissions be forecasted from these indicators across different development contexts?
2. Can countries be reliably classified into emissions tiers or energy transition stages from their structural characteristics? Which features drive those classifications?

These questions can help inform climate and energy policies by identifying which structural factors most strongly drive emissions disparities.

Data Sources & Description

The dataset used in this project will be the Global Data on Sustainable Energy (2000–2020) found on Kaggle (Tanwar, 2023). It contains 3,649 rows and 21 columns, with one row per country per year across approximately 175 countries. Key variables include: access to electricity (% of population), clean fuels for cooking, renewable energy share (%), electricity from fossil fuels and renewables (TWh), low-carbon electricity (%), primary energy consumption per capita (kWh/person), CO_2 emissions (kt), GDP per capita, and GDP growth. Two columns (financial flows to developing countries and renewables as a share of primary energy) are approximately 40% missing and will therefore be further evaluated prior to analysis.

Methodology

The dataset will first be preprocessed to ensure it is clean and suitable for analysis. This includes correcting data types, handling missing values, and normalizing variables. New variables may also be derived from existing ones where the raw data alone may not provide the information required for analysis (for example, generating category labels).

Two supervised machine learning approaches will then be applied. The first is regression, which will be used to predict a country's per-capita CO_2 emissions from its energy, economic, and infrastructure characteristics. The second is classification, where countries will be assigned to categories such as low, medium, or high emitter, based on those same characteristics. Multiple model types will be tested and compared to identify which performs best. For both approaches, the data will be split into training and testing sets to evaluate performance with unseen observations.

References

Tanwar, A. (2023). Global.Data.on.Sustainable.Energy.(8666_8686).[Dataset]. Retrieved from Kaggle: https://doi.org/10.34740/kaggle/dsv/6327347

United Nations. (2025). Retrieved from https://sdgs.un.org/goals/goal7#progress_and_info